

Advanced Quantitative Text Analysis

Day 3: Unsupervised Learning, Topic Models, and Scaling

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Yesterday and Today

- **Yesterday:** concepts, labels, and supervised classification
 - dictionaries
 - annotation
 - codebooks
 - supervised machine learning
 - validation and error analysis
- **Today:** what if we do not have labels?
 - clustering
 - topic models
 - structural topic models
 - text scaling
 - validation without ground truth

Today

- ① Supervised vs. unsupervised learning
- ② Measurement without labels
- ③ Clustering
- ④ Topic models
- ⑤ LDA and STM
- ⑥ Text scaling and Wordfish
- ⑦ Validation and interpretation
- ⑧ Practical session

The Main Problem

Unsupervised models discover structure.

They do not automatically discover meaning.

Meaning comes from interpretation, theory, and validation.

Machine Learning Paradigms

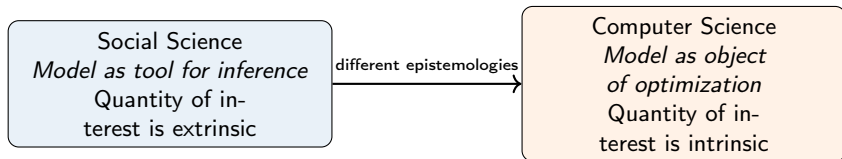
Supervised

- outcome variable defined
- labels available
- objective function clear
- evaluate against held-out labels

Unsupervised

- no predefined labels
- structure discovery
- objective is model-specific
- validation is harder

Measurement and Paradigms



What Is Measurement?

- Measurement = assigning numbers or symbols to phenomena according to rules.
- Three linked steps:
 - **Conceptualization:** What do we want to capture?
 - **Operationalization:** How can we represent it empirically?
 - **Quantification:** How do we express it numerically?
- Measurement maps theoretical constructs onto observed data.

Measurement in Social Science

- Measurement is often **theory-driven**.
- The goal is to capture latent constructs:
 - ideology
 - polarization
 - framing
 - attention
 - uncertainty
- Accuracy means correspondence between the measure and the theoretical concept.
- The model is a means to an end.

Focus: construct validity and interpretability.

Measurement in Computer Science

- Measurement is often **optimization-driven**.
- The model produces measurable quantities:
 - loss
 - accuracy
 - error
 - likelihood
 - reconstruction quality
- Accuracy means optimizing the model's objective.
- The model can become the object of interest.

Focus: predictive performance and optimization.

What Is an Objective Function?

- The objective function defines what the model tries to achieve.
- It translates a goal into something computable.
- It is the model's definition of success.

Model	Objective	Success means
Regression	minimize squared error	accurate predictions
K-means	minimize within-cluster variance	compact clusters
PCA	minimize reconstruction error	efficient compression
LDA	maximize likelihood	probable word-topic structure
Wordfish	maximize likelihood	plausible latent positions

Objective Function vs. Quantity of Interest

Objective function

- internal to the model
- mathematically optimized
- usually explicit
- defines model success

Quantity of interest

- external to the model
- theoretically motivated
- often latent
- defines research success

The model can optimize perfectly and still measure the wrong thing.

Why the Objective Function Matters

- It defines what the model cares about.
- Different objectives reveal different structures.
- If the objective does not align with theory, results can mislead.
- In unsupervised learning, this is especially dangerous because there are no labels to correct us.

Good measurement

Alignment between the theoretical quantity of interest and the model's objective.

What Does $\hat{\theta}$ Mean?

- Every model estimates parameters:

$$\hat{\theta}$$

- These may be:
 - regression coefficients
 - cluster centroids
 - topic proportions
 - word-topic probabilities
 - document positions
 - embeddings
- But the meaning of $\hat{\theta}$ depends on the research paradigm.

$\hat{\theta}$ in Two Paradigms

Computer Science

$$\hat{\theta} = \arg \min_{\theta} L(\theta)$$

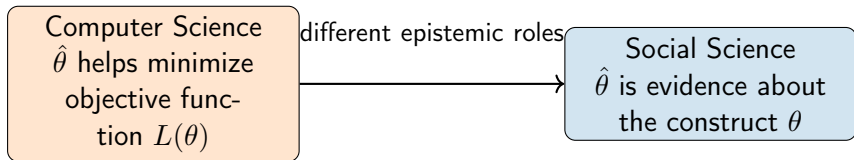
- parameter estimates help optimize performance
- interpretation may be secondary
- $\hat{\theta}$ is often a means to prediction

Social Science

$$\hat{\theta} \approx \theta$$

- parameter estimates are evidence about the world
- interpretation is central
- $\hat{\theta}$ approximates a theoretical construct

Two Roles of $\hat{\theta}$



Same symbol, different inferential meaning.

When There Is No True θ

$$L(X, \hat{S})$$

In unsupervised learning, the model does not minimize distance between a true parameter θ and an estimate $\hat{\theta}$.

Instead, it searches for an internal structure $\hat{S}$ that organizes the observed data X .

K-means:	compact clusters
PCA:	low-dimensional reconstruction
LDA:	latent topic-word structure
Wordfish:	latent document positions

The Problem

- Machine learning measures patterns, not constructs.
- Models optimize statistical objectives, not theoretical meanings.
- Without a mapping between theory and model, $\hat{S}$ may have no interpretable referent.
- Therefore, unsupervised models require an additional interpretive step.

Core problem

What does the discovered structure $\hat{S}$ correspond to in the world?

From Truth to Discovery

Truth-oriented: $\theta \rightarrow \hat{\theta}$

Model approximates a known or assumed data-generating process.

Discovery-oriented: $X \rightarrow \hat{S}$

Model identifies patterns or latent structures in the data.

In unsupervised learning, meaning is not recovered automatically; it is constructed through interpretation.

Epistemology of Discovery

Measurement as Mapping

$$\theta \longrightarrow X \longrightarrow \hat{\theta}$$

Model represents an existing theoretical construct.

Measurement as Modeling

$$X \longrightarrow \hat{S} \longrightarrow \text{interpretation}$$

Model induces structure that must be interpreted.

Families of Unsupervised Methods

Family	Goal	Examples
Clustering	group similar documents	k-means, hierarchical clustering
Topic modeling	find latent themes	LDA, STM, NMF
Scaling	place texts on dimensions	Wordfish, Wordscores
Embedding-based discovery	group by semantic similarity	BERT embeddings, BERTopic

Clustering

- Goal: group similar documents together.
- Input: document vectors.
- Output: clusters.
- Similarity can be based on:
 - raw counts
 - TF-IDF
 - cosine similarity
 - embeddings

K-Means Clustering

- Partitions data into K non-overlapping clusters.
- Each observation belongs to exactly one cluster.
- Clusters are collectively exhaustive:

$$C_1 \cup C_2 \cup \dots \cup C_K = \{1, \dots, n\}$$

- Clusters are non-overlapping:

$$C_k \cap C_{k'} = \emptyset \quad \text{for } k \neq k'$$

K-Means Objective Function

$$\min_{C_1, \dots, C_K} \sum_{k=1}^K \sum_{i \in C_k} \|x_i - \mu_k\|^2$$

$$\mu_k = \frac{1}{|C_k|} \sum_{i \in C_k} x_i$$

- μ_k = centroid of cluster C_k
- Intuition: assign points so each cluster is as compact as possible.

K-Means Algorithm

- ① Choose number of clusters K .
- ② Randomly initialize cluster centroids.
- ③ Assign each document to the nearest centroid.
- ④ Recompute centroids.
- ⑤ Repeat until assignments stabilize.

Warning

K-means converges to a local optimum, not necessarily the global optimum.

Choosing the Number of Clusters

- **Elbow method:** plot within-cluster sum of squares against K .
- **Silhouette score:** compare distance to own cluster vs. other clusters.
- **Theory-informed choice:** choose K based on expected structure.
- **Interpretability:** choose a solution that can be explained and validated.

Hierarchical Clustering

- Builds nested groupings of documents.
 - Does not require choosing K immediately.
 - Often visualized as a dendrogram.
 - Useful for exploratory analysis.
-
- Agglomerative: start with documents, merge clusters.
 - Divisive: start with all documents, split clusters.

Clustering: Validation Questions

- Are documents within clusters similar?
- Are clusters distinct from each other?
- Are clusters stable across preprocessing choices?
- Do clusters correspond to interpretable categories?
- Do clusters predict external metadata?
- Are clusters substantively useful?

Topic Models

- Goal: discover latent themes or topics in a collection of documents.
- Topic modeling is a family of models, not a single algorithm.
- Common approaches:
 - Latent Dirichlet Allocation
 - Structural Topic Model
 - Non-negative Matrix Factorization
 - embedding-based topic discovery

What Is a Topic?

- In topic models, a topic is not a label.
- A topic is usually a probability distribution over words:

$$\phi_k = P(w \mid z = k)$$

- High-probability words help us interpret the topic.
- The researcher gives the topic a substantive label.

Documents as Mixtures of Topics

- A document can contain multiple topics.
- Each document has topic proportions:

$$\theta_d = (\theta_{d1}, \theta_{d2}, \dots, \theta_{dK})$$

$$\sum_{k=1}^K \theta_{dk} = 1$$

- Example:
 - 50% economy
 - 30% migration
 - 20% party politics

Topics as Word Distributions

Topic	Top word 1	Top word 2	Top word 3
Economy	inflation	prices	unemployment
Migration	border	asylum	refugees
Climate	emissions	energy	warming
Health	hospitals	patients	doctors

The model gives word distributions; the researcher interprets them.

Mixture Models

- Mixture models assume observations come from multiple latent components.
- Each component has its own distribution.
- Each observation has a probability of belonging to each component.

$$P(x_i) = \sum_{k=1}^K \pi_k P(x_i \mid z_i = k)$$

In text

A document can be a mixture of topics.

Latent Dirichlet Allocation

- Bayesian generative model for discovering topics in text.
- Each document is modeled as a mixture of topics.
- Each topic is modeled as a distribution over words.
- Documents are soft mixtures of topics, not hard clusters.

LDA Generative Story

- ① For each topic k , draw a distribution over words:

$$\phi_k \sim \text{Dirichlet}(\beta)$$

- ② For each document d , draw a distribution over topics:

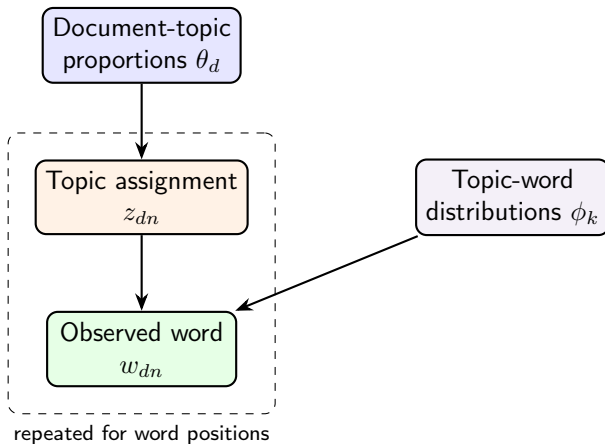
$$\theta_d \sim \text{Dirichlet}(\alpha)$$

- ③ For each word position n in document d :

$$z_{dn} \sim \text{Categorical}(\theta_d)$$

$$w_{dn} \sim \text{Categorical}(\phi_{z_{dn}})$$

LDA Diagram



What LDA Estimates

- Topic-word distributions:

$$P(w \mid \textit{topic})$$

- Document-topic proportions:

$$P(\textit{topic} \mid \textit{document})$$

- Topic assignments for words:

$$P(\textit{topic} \mid \textit{word}, \textit{document})$$

- A lower-dimensional representation of documents.

Dirichlet Distribution

- A distribution over probability vectors.
- It describes random vectors that sum to 1:

$$\boldsymbol{\theta} = (\theta_1, \dots, \theta_K), \quad \sum_{k=1}^K \theta_k = 1$$

- Parameters control concentration.
- Low values: sparse distributions.
- High values: diffuse distributions.

Dirichlet in LDA

Variable	Drawn from	Meaning
θ_d	Dirichlet(α)	topic distribution for document d
ϕ_k	Dirichlet(β)	word distribution for topic k

- α : how many topics appear per document.
- β : how many words dominate each topic.

Choosing the Number of Topics

- The number of topics K is usually chosen by the researcher.
- There is no universally correct K .
- Too few topics:
 - broad, mixed, hard to interpret
- Too many topics:
 - fragmented, redundant, unstable

Interpreting Topics

For each topic, inspect:

- highest-probability words
- most exclusive words
- representative documents
- documents with high topic proportions
- topic correlations
- metadata patterns

Important

Do not label a topic from top words alone.

Semantic Coherence and Exclusivity

Semantic coherence

- do top words co-occur?
- are they internally consistent?
- often favors broad topics

Exclusivity

- are top words distinctive?
- do they separate this topic?
- often favors sharper topics

Good topics usually balance coherence and exclusivity.

Common Topic Model Failure Modes

- Topics capture source instead of issue.
- Topics capture language instead of theme.
- Topics capture time periods instead of substance.
- Topics split one concept into many fragments.
- Topics merge several concepts into one.
- Topics are driven by boilerplate or named entities.
- Topic labels become wishful thinking.

From LDA to STM

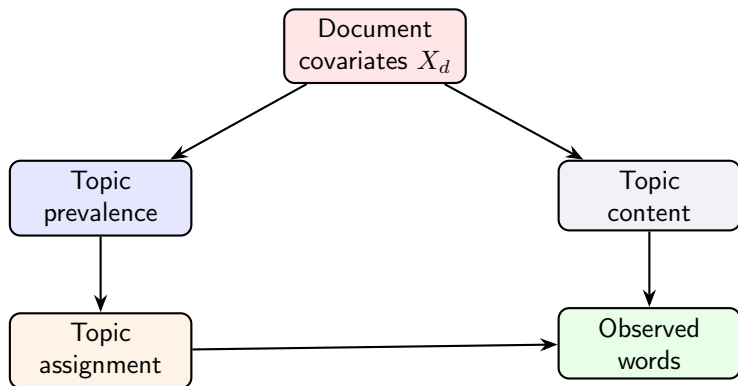
- **LDA:** all documents share the same prior structure over topics.
- **STM:** allows topic prevalence and topic content to vary with document covariates.
- Covariates can affect:
 - **Topic prevalence:** how much a topic appears.
 - **Topic content:** which words are used within a topic.

STM Covariates

Examples of document-level covariates:

- time
- source
- outlet type
- party
- country
- author gender
- platform
- government/opposition status

How STM Extends LDA



Covariates structure which topics appear and how they are expressed.

Topic Prevalence

- Topic prevalence asks:
 - Which documents talk more about a topic?
- Example questions:
 - Do tabloids discuss crime more than broadsheets?
 - Does climate coverage increase after extreme weather events?
 - Do parties emphasize different issues?
 - Does topic attention change over time?

Topic Content

- Topic content asks:
 - Do groups use different words when discussing the same topic?
- Example:
 - Topic: migration
 - Group A: border, illegal, crisis
 - Group B: asylum, protection, integration
- Same topic, different framing.

STM Warnings

- Covariate effects are not automatically causal.
- Topic labels are still researcher interpretations.
- Results depend on K , preprocessing, and initialization.
- Highly correlated covariates complicate interpretation.
- Content covariates can be difficult to interpret.

Text Scaling

- Goal: place documents or actors along a latent dimension.
- Often used for:
 - ideology
 - policy positions
 - conflict dimensions
 - government–opposition positioning
- Unlike topic modeling, scaling usually assumes a dominant dimension.

Topic Models vs. Scaling

Topic model	What themes are discussed?
Scaling model	Where are documents or actors located on a latent dimension?

- Topic models: mixed themes.
- Scaling models: positions on dimensions.

Wordfish: Intuition

- Wordfish estimates document positions from word counts.
- Documents have latent positions.
- Words have discrimination parameters.
- Words used more by one side help define the dimension.

Wordfish Model

$$y_{ij} \sim \text{Poisson}(\lambda_{ij})$$

$$\log(\lambda_{ij}) = \alpha_i + \psi_j + \beta_j \omega_i$$

- y_{ij} : count of word j in document i
- α_i : document fixed effect
- ψ_j : word fixed effect
- β_j : word discrimination
- ω_i : latent document position

Assumptions of Text Scaling

- Texts vary along a meaningful latent dimension.
- Word usage reflects position on that dimension.
- Word meanings are stable across texts.
- Documents are comparable.
- Variation in word use reflects position, not just topic.
- Scale origin and direction are arbitrary.

Scaling Failure Modes

- The dominant dimension is not the intended dimension.
- Positions reflect topics rather than ideology.
- Positions reflect document length or genre.
- The scale shifts over time.
- Multiple dimensions are collapsed into one.
- Interpretation depends too heavily on a few words.

Why Validate Unsupervised Models?

- No ground truth means models can always find patterns.
- Validation checks whether discovered structure reflects:
 - the data
 - the world
 - the theory
 - or merely the algorithm
- Without validation, unsupervised results are just model artifacts.

Validation Strategy Matrix

Strategy	Question
Internal metrics	Does the model optimize its own objective well?
Interpretability	Can humans make sense of the output?
Stability	Do results persist across seeds, samples, and pre-processing?
External validation	Do outputs correspond to metadata or expert knowledge?
Predictive validation	Do outputs predict relevant outcomes?
Construct validation	Does this measure capture the intended concept?

Validating Topic Models

- **Semantic validity:** do top words form coherent themes?
- **Convergent validity:** do topic proportions correlate with external variables or manual coding?
- **Quantitative metrics:**
 - topic coherence
 - held-out likelihood
 - perplexity
- **Human-in-the-loop validation:**
 - experts or coders rate interpretability
 - word intrusion
 - topic intrusion

Stability Checks

- Re-estimate with different random seeds.
- Try different numbers of topics or clusters.
- Try alternative preprocessing choices.
- Refit on subsamples.
- Compare results across time periods or sources.
- Inspect whether substantive conclusions change.

External Validation

Compare model output to variables not used by the model:

- party family
- media outlet type
- election period
- known issue ownership
- expert ideology scores
- hand-coded labels

Best case

The validation variable was not used to tune interpretation.

Confirmation Bias

- Unsupervised models invite interpretation.
- Interpretation can become confirmation bias.
- Risks:
 - labeling topics according to what we hoped to find
 - ignoring unstable or inconvenient topics
 - overinterpreting vague word lists

Good practice

Report ambiguity, instability, and alternative interpretations.

Reporting Unsupervised Results

Report:

- corpus and preprocessing
- representation
- model and tuning choices
- number of topics/clusters/dimensions
- criteria for choosing the model specification
- top words and representative documents
- validation checks
- robustness checks
- limitations of interpretation

Estimate, interpret, and validate unsupervised models

Practical Session: Tasks

- ① Load the corpus.
- ② Create a document-term matrix.
- ③ Estimate a topic model.
- ④ Inspect top words.
- ⑤ Inspect representative documents.
- ⑥ Compare models with different numbers of topics.
- ⑦ Relate topics to metadata.
- ⑧ Write a short validation memo.

Topic Interpretation Template

For each topic:

- Top probability words:
- Most exclusive words:
- Representative documents:
- Tentative label:
- Alternative labels:
- Evidence supporting the label:
- Evidence against the label:
- Possible artifact:

Compare Models

Estimate at least two models:

Model	Question
$K = 10$ topics	Are topics too broad?
$K = 20$ topics	Are topics interpretable?
$K = 40$ topics	Are topics too fragmented?
Alternative preprocessing	Are results stable?

Validation Memo

Write 5–7 sentences:

- ① What model did you estimate?
- ② Which topic or cluster is most interpretable?
- ③ What evidence supports your interpretation?
- ④ What evidence weakens it?
- ⑤ What validation check would you do next?

Summary: Comparing Unsupervised Methods

	Goal	Output	Example
Clustering	grouping	cluster labels	K-means
Topic modeling	thematic discovery	topic-word probabilities	LDA / STM
Scaling	latent dimension	position scores	Wordfish

Key Takeaways

- Unsupervised methods discover structure; they do not automatically discover concepts.
- Clustering groups similar documents.
- Topic models represent documents as mixtures of themes.
- STM connects topics to document metadata.
- Scaling places documents or actors on latent dimensions.
- The output must be interpreted and validated.
- Stability, external validation, and artifact inspection are essential.

The Main Lesson

Unsupervised learning is not label-free measurement.

It is structure discovery.

The measurement happens when we connect structure back to concepts, evidence, and validation.

Tomorrow

- Embeddings
- Transformers
- Contextual representations
- Multilingual analysis
- Large language models
- Prompting and zero-shot classification